

REVIEW

Comparative Analysis of Virus-Induced CRISPR Therapy vs. RNAi Gene Regulation for Combating Cancer-Linked Gene Expression

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I. Abstract

Advancements in gene therapy over the last two decades have shown that this type of therapy could be a viable solution for cancer. Two promising gene-modulation strategies so far are CRISPR-Cas9 editing and RNAi-based gene silencing that work to combat cancer. Hence, in this literature review, we evaluate and contrast CRISPR and RNAi across efficacy, safety, and practical viability as anticancer therapeutics. To do this, databases like PMC and Nature were scanned using keywords based around “CRISPR,” “RNAi,” and “Cancer” to evaluate the following criteria: we tried to find the off-target effects of both tools, cancer resistance, tumor size effects, and the feasibility of CRISPR and RNAi. Key findings were that CRISPR usually had riskier but more permanent effects, while RNAi offered more risks of off-targeting but had less permanent effects. In terms of efficacy, CRISPR may cause more durable cancer gene changes, while RNAi requires sustained dosing. While both technologies are relatively new, based on the findings, we concluded that there is active potential for these technologies to serve as promising therapies individually or combined. As delivery systems and manufacturing costs continue to improve, CRISPR-Cas9's specificity and permanence make it likely to surpass RNAi as the primary gene-modulation strategy in oncology. However, this comparative analysis can help guide clinicians and researchers to consider combinatory tools that use the two mechanisms to offset each other's limitations, and to thoroughly assess the risks of each therapy.

II. Introduction

Clustered regularly interspaced short palindromic repeats, also known as CRISPR, is a gene-editing tool that can be used to battle diseases, such as cancer. CRISPR works using proteins called CRISPR-associated proteins (Cas proteins) that may serve as cleavage enzymes within a section of the genome that needs editing, such as mutations that cause diseases. These Cas proteins are brought to the target gene site with the help of Guide-RNA. Guide-RNA's job is just as its name implies: it leads Cas proteins to the gene sequence to be edited, where the Cas enzyme binds to that sequence, resulting in a double stranded break of DNA. Afterwards, this damage to the cell's genetic material signals the cell's repair mechanism to start editing DNA

sequences. Two different methods of repair can subsequently take place: homology-directed repair reforms DNA by utilizing the homologous chromosome as a template; non-homologous end-joining refers to the adhesion together of broken ends of DNA. (Raffi et al., 2022; Wang et al., 2023). HDR is the preferred method in this review as it is able to correct the mutation rather causing more genomic damage.

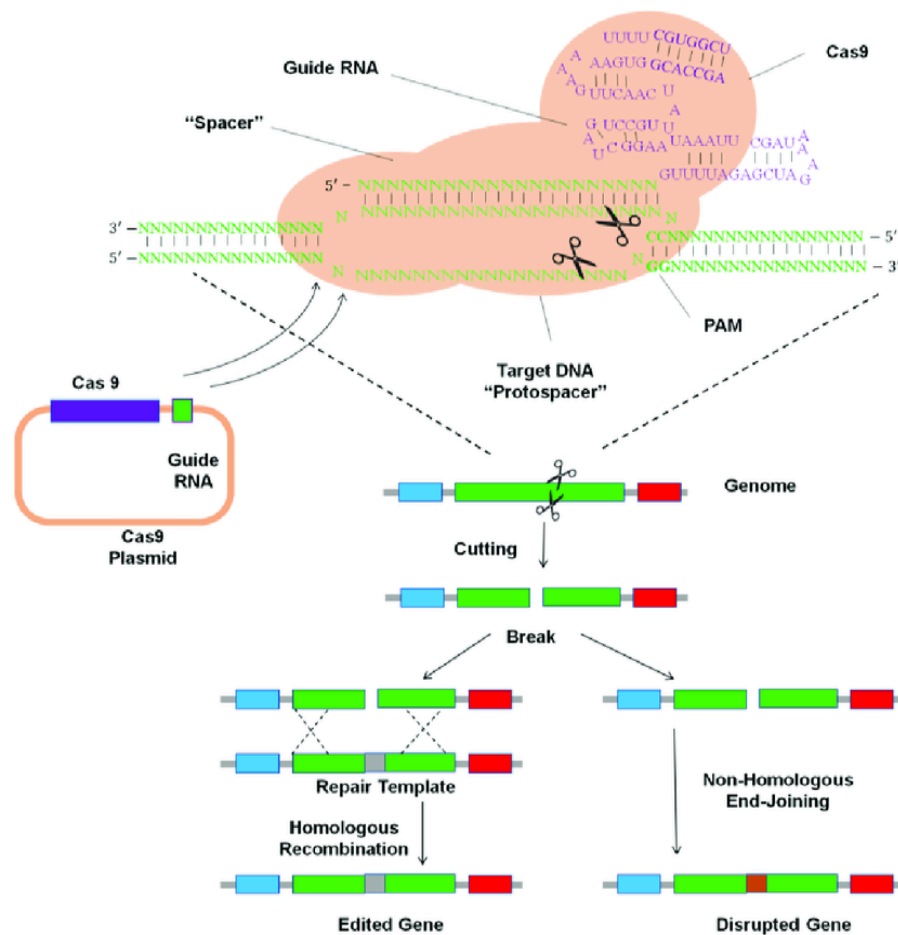


Figure 1: Illustration of gRNA matching with sequence, Cas9 cleavage, and potential for homology directed repair or non-homologous end-joining to result in genomic editing
Adopted from: Upadhye J V, Noor Shah U, Mudhol B. The Possibilities of Gene Drives for Managing Populations and Controlling Diseases. *Salud, Ciencia y Tecnologia*. 2023;3(S1):451. <https://doi.org/10.56294/saludcyt2023451>

RNA-interference, also known as RNAi, is a method to decrease gene expression that works post-transcriptionally. This process occurs with the help of ncRNA (non-coding RNA), specifically small interfering RNA (siRNA) and microRNA (miRNA), which are loaded into a structure called the RNA-induced silencing complex (RISC). This then causes the targeted mRNA transcript to break down before it can be translated into a protein (Tian et al., 2021).

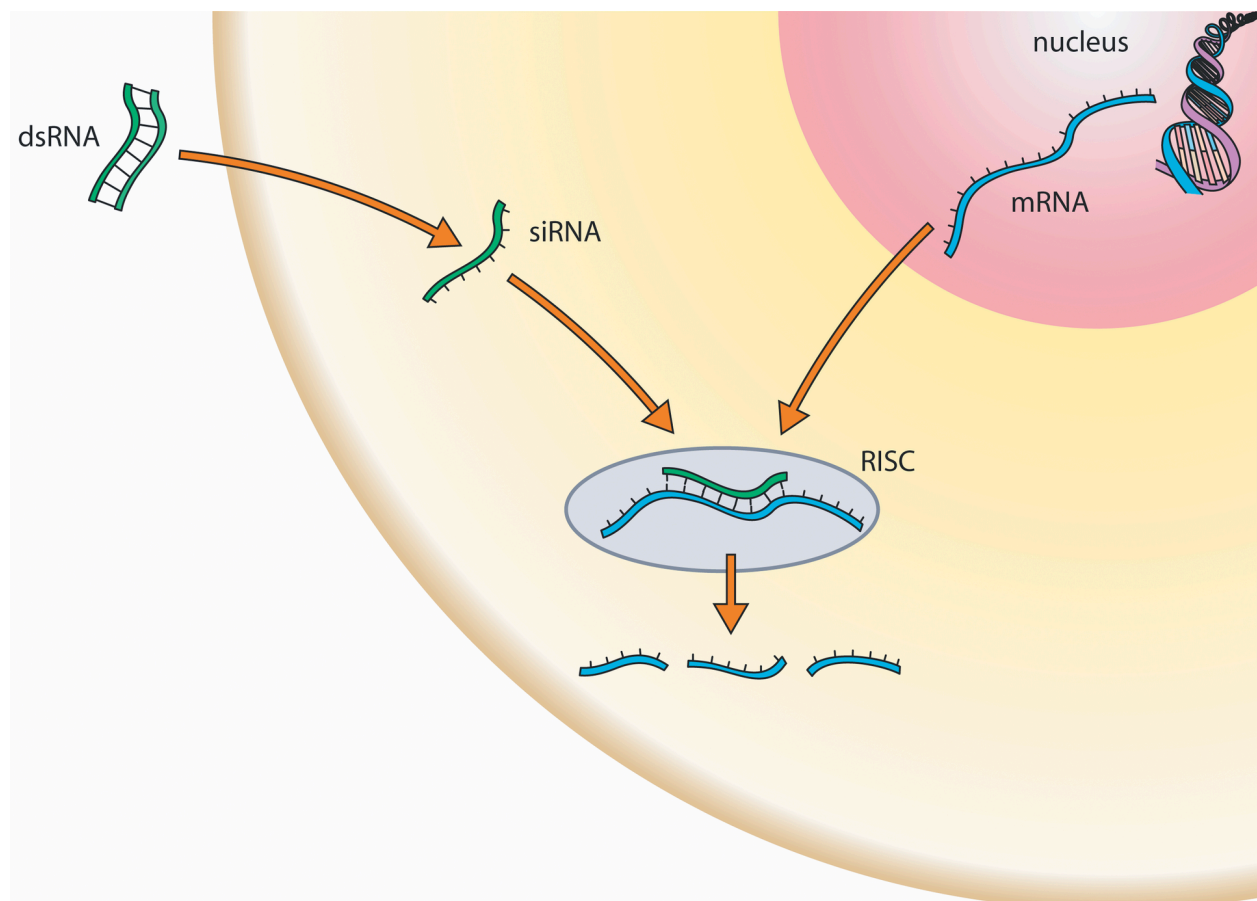


Figure 2: siRNA binds to RNA-induced silencing complex, which blocks mRNA transcript from encoding translated proteins. This “knocks-down” the specific gene encoded in the mRNA.

Adopted from: Robinson, R. (2004). RNAi Therapeutics: How Likely, How Soon? *PLoS Biology*, 2(1), e28. <https://doi.org/10.1371/journal.pbio.0020028>

Cancer is broadly defined as a group of diseases characterized by the uncontrolled growth and division of abnormal cells because of mutations that affect the cell cycle (Hanahan & Weinberg, 2011). Existing cancer treatment, such as chemotherapy, may cause additional side effects due to an inability to target specific problematic cells. Therefore, minimally-invasive treatments that are able to address specific genomic mutations associated with cancer removal are of prime interest. T cells, a central component of adaptive immunity, are capable of recognizing and eliminating malignant cells through antigen-specific mechanisms as well (Cassioli & Baldari, 2022). Currently, the two therapeutics mentioned above are viable ways to treat cancer by either knocking down protein translation (RNAi) or directly correcting the gene mutation (CRISPR). In this paper, we compare the technologies of RNAi and CRISPR in addressing the expression of cancer-linked mutations through a comparative analysis of existing literature and conclude that both technologies have a high therapeutic potential.

III. Methodology

Identifying relevant sources for this research consisted of doing literature search using databases including PubMed Central (PMC), Nature, Frontiers, SpringerLink, and ScienceDirect. Combinations of keywords put in search engines include, “CRISPR cancer,” “CRISPR-Cas9 cancer therapy,” “CRISPR gene editing cancer,” “RNA cancer therapeutics,” “RNAi,” “RNA-based cancer treatment,” “CRISPR CAR-T therapy,” “CRISPR T-cell engineering,” and “CRISPR cancer immunotherapy.” We prioritized peer-reviewed research and review articles that addressed the relationship between RNA, CRISPR-based gene editing, cancer mechanisms, RNA interference, and potential therapeutic applications. We then reviewed the titles and abstracts to determine their relevance and examined the full articles when they contained information applicable to the research question. Collections of summarizations, notes, and annotations were taken for a number of individual sources. Sources were selected based on their relevance to topics such as CRISPR mechanisms, cancer-associated genes, T-cell and CAR-T engineering, RNAi-based therapeutics, and specific cancer applications.

Next, databases were selected for their strength in medical-related papers and frontiers for different cancer-related therapies. Keywords were chosen to capture delivery, mechanistic, and outcome studies so that CRISPR and RNAi could be evaluated on equivalent metrics. The comparison criteria are as follows:

- Frequency of risk/off-target effects
- Resilience of cancers
- Tumors
- Feasibility

Off-targeting is defined as the unintended CRISPR-Cas9 cleavage at genomic locations other than the intended target, and can occur because of partial sequence homology between the guide RNA and non-target DNA regions. It is found in the literature through whole-genome sequencing detection rates, typically reported as a percentage of total edits. In this paper, it will be examined under the lens of off-target effects, and which tool has a lower likelihood of such an event is considered superior to the other therapy for this reason.

Resilience of a cancer is its ability to come back once CRISPR and/or RNAi have already operated on the same cancer. Oftentimes, patients who receive treatments like chemotherapy get cancer again without warning, meaning that these tools should be evaluated on their ability to suppress the cancer from coming back. With literature, these sources mainly come as case-studies or new reports, as CRISPR/RNAi are relatively new technologies.

Next, tumors are the abnormal accumulation of tissue and cells when cancer is found. In the context of RNAi and CRISPR, discussion will follow the types of cancer and tumors that each of

the tools attack, respectively. In literature, this is typically reported as cancers that have been removed effectively in regards to a specific tool.

Feasibility is defined as the practicality of a product, and its ability to work at larger scales. Here, we compare the delivery method of CRISPR and RNAi, and some of their drawbacks. These figures are not yet quantified in literature, so in our paper we will evaluate the feasibility of the drug for human use. In literature, these studies are found as reports or clinical trials.

IV. Results

A. RNAi Technology

RNA interference or RNAi is a gene silencing technology that can silence or inhibit genes and has a 60-80% success rate when silencing specific genes (Abdelrahim et al., 2006; Munkácsy et al., 2016). The mechanism behind RNAi starts with the Dicer enzyme recognizing double stranded RNA (dsRNA) which is usually where diseases like cancer tend to be integrated (Abdelrahim et al., 2006; Fu et al., 2023; Lv et al., 2006). Once Dicer processes dsRNA, it cuts it into approximately 21-23 nucleotides in length with an overhang of 3' (Abdelrahim et al., 2006; Fu et al., 2023; Mansoori et al., 2014; Tiemann et al., 2009; Yoon et al., 2022). Next, siRNA, which is in Argonaute found in the RISC protein separates dsRNA into the passenger strand, which is discarded, and the guide strand, which pairs with mRNA (Mansoori et al., 2014; Tiemann et al., 2009; Wucherpennig et al., 2016). In the situation of an exact or near perfect match, siRNA pairs to mRNA and cuts the targeted gene (Lv et al., 2006; Tian et al., 2021; Tiemann et al., 2009). On the other hand, when it is a partial match, the miRNA pairs with the mRNA and reduces or limits the targeted gene's transcription (Fu et al., 2023; Mansoori et al., 2014; Tian et al., 2021; Yoon et al., 2022). siRNA and miRNA are both crucial parts of silencing cancer cells through removing and reducing. Scientists have made developments combining RNAi technology with nanoplatforms and nanomaterials. In addition, these two developments can be implemented in improving diagnoses and immunotherapies (Xu et al., 2026; Yoon et al., 2022). Moreover, a disadvantage of RNAi is that it may cause off-target effects (OTE) when binding to mRNA fails, which is an important aspect of this treatment that scientists are currently trying to reduce (Champagne et al., 2019; Fu et al., 2023; Fujita et al., 2013; Huang et al., 2020; Tian et al., 2021; Tiemann et al., 2009; Wucherpennig et al., 2016). TAG-RNAi is an advancement of RNAi that avoids and limits OTE (Champagne et al., 2019).

B. CRISPR Technology

CRISPR's cancer applications rely on targeted genetic modification to change pathways which have an influence on tumor growth or antitumor immune activity. For example, in T-cell cancer therapy researchers used Cas9 and sgRNAs to disrupt the PD-1 gene in primary human T cells. PD-1 is an inhibitory immune checkpoint, and the study found that disrupting it reduced PD-1 expression, while keeping T-cell proliferation during extended cultivation. (Ravichandran & Maddalo 2023; Su et al., 2016; Wang et al., 2022)

The modified T cells produced more IFN- γ when exposed to tumor-associated antigens which include antigens from melanoma and gastric cancer patients and showed enhanced cytotoxicity against tumor-cell lines (Su et al., 2016). This illustrates the therapeutic mechanism of using CRISPR to remove an inhibitory signal from immune cells. This can potentially allow them to grow a stronger response against cancer. The same principle can be applied to CAR-T therapy where CRISPR-based engineering can modify T cells while introducing therapeutic CARs. The CLASH platform used Cas12a/Cpf1-mediated editing and targeted knock-in to generate and screen different CAR-T cell variants. This ultimately identified a PRDM1 exon 3-skipping modification associated with increased proliferation, stem-like and central-memory characteristics, and persistence of CAR-T cells in cancer models (Dai et al., 2023; Wellhausen 2022). CRISPR can therefore be applied therapeutically not only to alter immune checkpoints such as PD-1 but also to optimize engineered immune cells for characteristics such as persistence and antitumor activity. Simultaneously, other RNA-based approaches discussed in the literature can target cancer-associated genes through mRNA, siRNA, miRNA, antisense oligonucleotides, and CRISPR-related strategies.

CRISPR has been employed in clinical trials to treat various cancer types. Specifically, the use of CRISPR and Chimeric Antigen Receptor (CAR)-T cells has shown vast potential and success in cancer treatments. CAR-T therapy, the genetic engineering of T cells to release CARs, allows T cells to bind to antigens on cancer cells, activating the T cells to kill the cancer cells. CAR-T therapy has recently developed as a technique for treating blood-related cancers such as B cell leukemia. Furthermore, clinical trials have occurred with CRISPR in treating leukemia, lymphoma, and multiple myeloma (Martinez & Moon, 2019; Pont et al., 2023). Recent research further demonstrated the success of utilizing CRISPR to target solid tumors, including those that occur in triple negative breast cancer (TNBC). TNBC, the most common type of breast cancer, caused by insertion and deletion in the TP53 gene or alterations in the Breast Cancer 1/2 (BRCA1/2) gene, Phosphatase and tensin homolog (PTEN), Phosphoinositide-3-kinase catalytic subunit (PI3KCA), MYC, Epidermal growth factor receptor (EGFR), Engrailed 1 (EN1), and Forkhead box C1 (FOXC1), among others. With assistance from lipid nanoparticles (LNPs) in a groundbreaking clinical study, researchers have been able to ensure targeted CRISPR delivery while mitigating OTEs to solid tumors. This LNP-mediated CRISPR therapy is being actively explored in breast cancer and other solid tumor-based cancers, including lung cancer, colorectal cancer, and hepatocellular carcinoma (Wang et al., 2024).

C. Comparison

Though CRISPR and RNAi are both successful at inhibiting cancer cells, an important caveat to keep in mind are their off-target effects (OTE). Research has shown that CRISPR causes less OTE compared to other technologies such as ZFNs and RNAi which may play an important role when determining treatment options (Wu, 2022). This is exemplified through the development of

CRISPR that targets the PD-1 gene that showed no OTE (Chamberlain et al., 2022). Many sources also highlight that RNAi's OTE is one of the biggest issues when using RNAi (Boettcher & McManus, 2015). Furthermore, the OTE of RNAi tends to shorten the duration of the therapy because of the technology losing efficiency (Tian et al., 2021). Overall, the majority of sources state that CRISPR is safer than RNAi in the context of OTE.

Furthermore, CRISPR and RNAi fundamentally differ in their resilience, mechanism target levels. CRISPR's mechanism for gene editing operates permanently on coding and non-coding regions of DNA, knocking out genes in cells that will never return to their previous genome (Azangou-Khyavy et al., 2020; C. Chen et al., 2023; Rafii et al., 2022; Wang et al., 2024; X. Chen et al., 2024; Zhang et al., 2024). In other words, if T cells are edited to be able to attack cancer cells, they will not return to their original, non-cancer-targeting forms, permanently halting the progress of metastasizing cancers in the body. On the other hand, RNAi targets already-transcribed mRNA in the cytoplasm through siRNA or miRNA. The gene knockdown that occurs due to RNAi, unlike that of CRISPR, does not alter the cell's genome; therefore, unlike CRISPR, the effects of RNAi are reversible (Boettcher & McManus, 2015; Ding et al., 2024; Schaible et al., 2022; Tian et al., 2021; Urak et al., 2023). This reversibility can be considered as both a benefit and drawback to RNAi; on one hand, there is no permanent damage undergone by the genome and thus less potential of everlasting harm. On the other, this impermanence could further the possibility of cancer recurrence.

Additionally, the types of tumors that each therapy excels with eliminating differ due to each therapy's mechanism basis. Unlike RNAi, CRISPR excels at immune cell modifications, as is exemplified by knocking out PD-1 receptors or the creation of specialized CAR-T cells through platforms like CLASH. Both therapies rely on LNPs and other nanomaterials to target and destroy solid tumors.

Feasibility is where the two technologies differ most in practice, because it depends on delivery, instead of the actual mechanism of the tools. CRISPR-Cas9 components are often delivered via viral vectors, which carries packaging size limits, risk of insertional mutagenesis, and immunogenicity concerns with repeated doses (Vaghari-Tabari et al., 2022). However, non-viral delivery methods for CRISPR are rapidly advancing as alternatives. Lipid nanoparticles (LNPs) can be injected directly into solid tumors, physically concentrating gene-editing components within the tumor site. One study achieved approximately 70% editing efficiency using intra-tumoral injection of Cas9 mRNA and sgRNA, though this approach currently works best for superficial or easily accessible tumors (Wu et al., 2026). LNP-encapsulated CRISPR/Cas9 has also been used for therapy in head/neck cancer models without viral components (Masarwy et al., 2025). On the other hand, RNAi molecules are relatively small and can be delivered non-virally through lipid nanoparticles; these approaches have been validated at commercial scale for liver-targeted diseases as well (Traber & Yu, 2023). This size and delivery advantage

makes RNAi easier to manufacture and administer repeatedly, but also means that RNAi's clinical feasibility is mainly concentrated in tissues that are easy to reach. However, CRISPR's viral delivery right now is more acute and can target very specific areas, meaning its current challenges can be fixed through nanoparticle systems designed for tumor targeting. With this, in terms of feasibility, because both are relatively new technology, they cannot be compared numerically, but both excel to address different manifestations of cancer in individual patients.

Overall, the general body of research on gene therapies applied to cancer treatments agrees that clustered regularly interspaced short palindromic repeats is slightly more effective and specific, but RNAi is still a powerful tool. CRISPR's permanent gene knockouts are a major advantage over RNA-interference, whose transient gene knock-downs are temporary and reversible. Additionally, the increased likelihood of off-target effects in RNAi therapy increases the risks that come with treatments, especially compared to the minimal OTE of CRISPR.

More research must be conducted to determine the most effective ways to utilize both CRISPR and RNA interference to combat cancer. Clinical trials should consider the uses of both therapies in adult organisms rather than cell cultures to better gauge the effects of these therapies on live cancer patients. Moreover, some researchers suggest a synergistic approach to using these therapies: CRISPR and RNAi can be used together such that phenotypes discovered with CRISPR can be validated with RNAi to rule out compensatory effects. This dual approach to cancer therapy has strengthened conclusions in drug discovery and functional genomics in past research, and must be investigated and discussed in greater depth in future research (Huang et al, 2020).

V. Conclusion

Taken together, this analysis indicates that a dual approach of CRISPR and RNAi is effective for cancer, based on the cases studied in this review. The main differentiation between the two technologies lay between off-target effects and whether the therapies were feasible for their respective targets. Looking ahead, continued innovation in tumor-targeted delivery systems (including lipid nanoparticles, exosome-based carriers, and engineered viral vectors with reduced immunogenicity) will likely determine how quickly CRISPR-based and RNAi therapies can become effective for large scale use. Combination approaches that pair CRISPR's permanent gene disruption with RNAi's lower-risk delivery infrastructure also need further investigation, as they may offer a practical middle path while CRISPR delivery technology improves. As both therapeutics continue to evolve, future clinical trials comparing direct outcomes across matched cancer types will be essential for determining which strategy, or combination of strategies, offers the greatest long-term benefit for patients.

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